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SARDAR PATEL UNIVERSITY**Master of Science (Information Technology) (Sem-I (NC)) Examination April - 2025****PS01CINT55: Operating System Concepts****Date: 17/04/2025 (Thursday)****Time: 10:00 A.M. to 01:00 P.M.****Total: 70 Marks****NOTE:** Figure to the right indicates full marks of the questions.**Q.1. Answer the following multiple choice questions.****[08]**

1. When can the binding of instructions and data to memory addresses be done?
 - A. Load Time
 - B. Compile Time
 - C. Execution Time
 - D. All of these
2. The two types of semaphores are
 - A. Decimal and Counting semaphores
 - B. Binary and Counting semaphores
 - C. Mutex and Counting semaphores
 - D. None
3. For which of the following purpose the Banker's algorithm is used?
 - A. Solving deadlock
 - B. Recover from deadlock
 - C. Preventing deadlock
 - D. None of these
4. Run time mapping from virtual to physical address is performed by
 - A. Scheduler
 - B. PCI
 - C. CPU
 - D. Memory Management Unit (MMU)
5. To access the services of operating system the interface is provided by the?
 - A. System Calls
 - B. Assembly Instructions
 - C. API
 - D. Library
6. When process try to access a page marked invalid causes a _____.
 - A. Lazy swapper
 - B. Page fault
 - C. Page Table
 - D. None of these
7. The time taken by a desired sector of the disk to rotate itself to the position where it can access the Read/Write heads is called. _____.
 - A. Seek Time
 - B. Bandwidth
 - C. Rotational Latency
 - D. Waiting time
8. Which of the type of OS reads and reacts in terms of actual time?
 - A. Time Sharing OS
 - B. Quick Sharing OS
 - C. Batch OS
 - D. Real Time OS

Q.2. Answer the following questions (Attempt any Seven).**[14]**

1. List the benefits of a Time-Sharing Operating System?
2. Define short term and long term schedulers.
3. Define Critical Section.

4. State the difference between process and program.
5. What is swapping?
6. Define Starvation.
7. List out different types of operations performed on a file.
8. Write definition of seek time in terms of disk scheduling.
9. What is address binding in OS?

Q.3. Answer the following questions.

- A. Discuss various types of Operating System (OS). [06]
- B. Differentiate between CLI and GUI. [06]

OR

- B. Discuss about Interrupt Handling. [06]

Q.4. Answer the following questions.

- A. Explain various states of Process in detail. [06]
- B. What is deadlock? Discuss Necessary Conditions for Deadlock. [06]

OR

- B. Discuss Representation of Process Scheduling using Queuing Diagram. [06]

Q.5. Answer the following questions.

- A. Define virtual memory. Discuss the role of virtual memory in detail. [06]
- B. Discuss about Contiguous Memory Allocation. [06]

OR

- B. List out various page replacement algorithms and explain any one with suitable example. [06]

Q.6. Answer the following questions.

- A. Discuss various File structures and access methods in detail. [06]
- B. List out various disk scheduling algorithm and explain any one algorithm of your choice. [06]

OR

- B. Write a Short note on RAID structures. [06]
